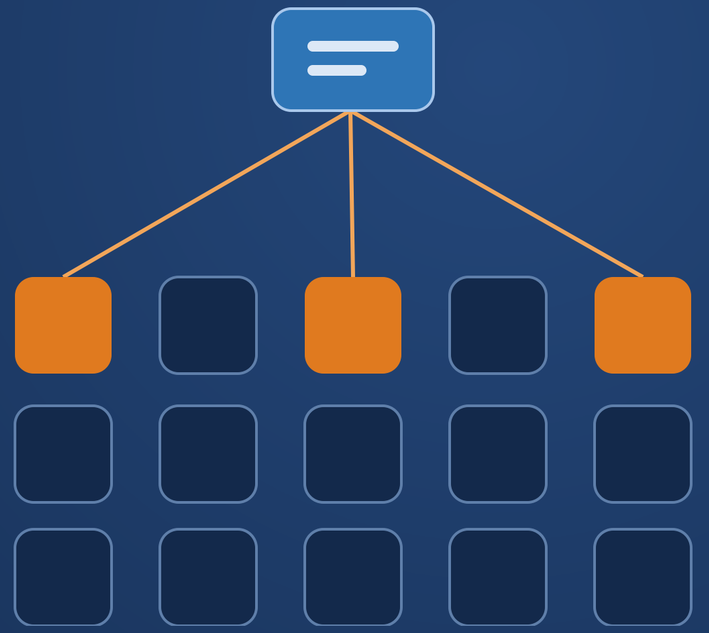
PROJECT PROPOSAL · DEPARTMENT OF CSE, BRAC UNIVERSITY

Enforcing booked slots on research-lab PCs

A self-hosted Headscale server that lets only the group in its slot reach its PC, piloted on 15 lab machines



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THE PROBLEM

Slots exist only on paper: anyone with a PC’s AnyDesk details can connect

At any time

AnyDesk connects through AnyDesk’s own servers. A PC’s ID and password work outside the slot too.

Into anyone’s work

They can stop another group’s training run, change its files or take over its desktop.

With no record

Nobody can tell afterwards who connected, or when.

Today each thesis group has two slots of about 12 hours a week on a lab PC, with one to five students, often on several devices each.

THE GOAL

The pilot makes the booking the rule: **only the group in its slot gets in**

> During a slot, only that group can reach its PC remotely. Outside it, nobody except the lab admins.
>
> Scope: 15 Linux lab PCs, one server in the lab, a group’s slots switched automatically.

BRACU sign-in only

Students use their BRACU Google account. The lab keeps no passwords.

The Sheet stays

The weekly routine stays in the lab’s Google Sheet, as today.

Everything logged

Every change and every enforcement step is recorded, then measured.

OPTIONS CONSIDERED

Of six ways to give remote access, **only Headscale is ours, free and policy-driven**

Option	Who runs it	Fit for slot-based access
AnyDesk only (today)	AnyDesk’s servers	No enforcement, no record of who connected
SSH forwarding or bastion	Ports opened on campus	SSH exposed to the internet; slots enforced on each PC
Plain WireGuard hub	Us; all traffic via the hub	No NAT traversal; keys and rules by hand per device
Tailscale (hosted)	Tailscale Inc., off campus	Free for 6 users, then $8 per user a month; data off campus
NetBird (self-hosted)	Us	Five services to run, and its own clients
Headscale (chosen)	**Us; direct peer-to-peer**	**One program and SQLite, official Tailscale apps, free (BSD-3)**

The deciding feature: the whole access policy is one document we rewrite and load at every slot change.

Tailscale pricing: tailscale.com/pricing (October 2026)

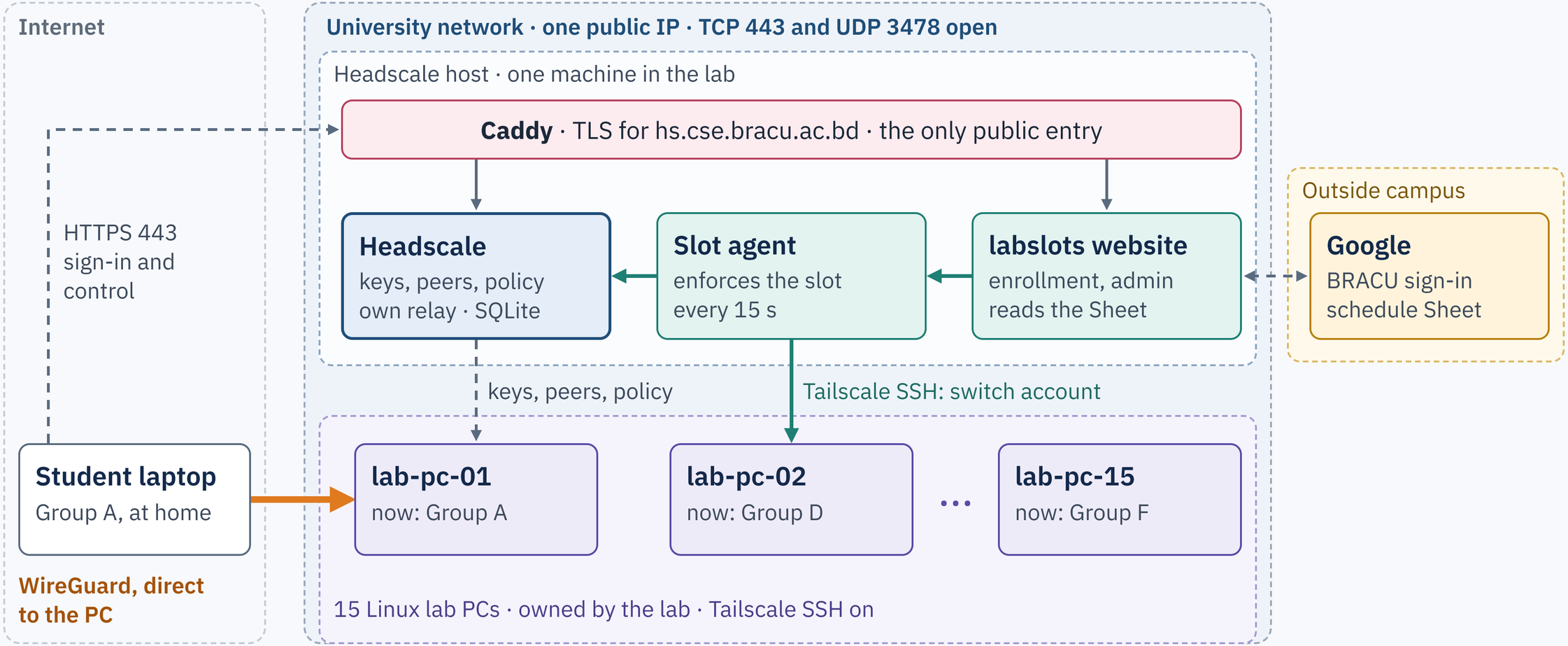
PART 1

How it works

The architecture, a connection, a slot change, who can reach what, the Sheet, the website and where it all runs

HOW IT WORKS · ARCHITECTURE

One machine in the lab decides who can reach which PC



student traffic (WireGuard) ------- control and sign-in (HTTPS) slot enforcement

HOW IT WORKS · A CONNECTION

A connection goes **direct when it can**, and through our own relay when it cannot

STEP 1

Sign in, get peers

Each device keeps an HTTPS session to Headscale and gets only the peers the policy allows.

STEP 2

Start on our relay

A new connection starts through our relay (DERP) while STUN finds each end’s public address.

STEP 3

Go direct if possible

If a direct UDP path works, traffic moves to it. If UDP is blocked, it stays relayed over TCP 443.

ALWAYS

Encrypted end to end

Every path is WireGuard. The relay only ever sees ciphertext; Headscale never sees traffic.

Our relay runs on the lab server, so relayed traffic stays on campus.

Off campus with only TCP 443 open, the rehearsal still connected at 27–37 ms.

After Tailscale, “How NAT traversal works” (2020)

HOW IT WORKS · EVERY SLOT CHANGE

Every 15 s, the slot agent makes the PCs match the schedule

Google Sheet
The lab’s weekly routine: a row per PC, a column per slot

→

labslots website
Reads it every 30 s, checks every cell, keeps the last good copy

→

Slot agent
Runs every 15 s, so a missed change is put right on the next run

→

1 · Slot ends, 17:55. The policy drops the outgoing group; its sessions and processes end.

2 · Next slot starts, 18:00. The incoming group is let in and its account made the only active one.

3 · On any error. Loads an admins-only policy: access fails closed.

One PC’s week: 14 slots, six groups (A–F), each with two; every slot ends five minutes before the next starts

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
06:00–17:55	**A**	**C**	**E**	**A**	**C**	**E**	*maintenance*
18:00–05:55	**B**	**D**	**F**	**B**	**D**	**F**	*spare*

HOW IT WORKS · WHO CAN REACH WHAT

Each group reaches **only its own PC, only in its slot**, only as its own account

Who	Can reach	Logs in as
Lab admins	every lab PC, any port	any non-root account
Slot agent	every lab PC, port 22	labsched only
Group in its slot	its own PC, port 22 and its ports	its group account only
Everyone else	nothing	nothing

No keys, no passwords

Tailscale SSH checks who is calling against the policy. Nothing to hand out, nothing to take back when a student leaves.

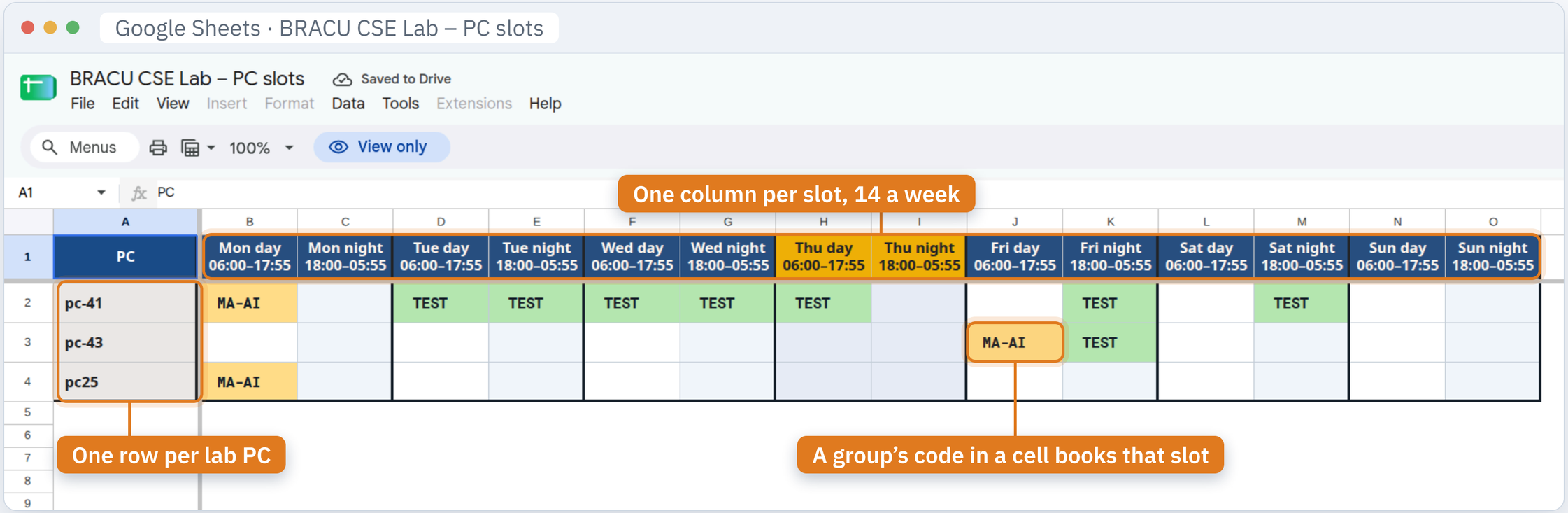
Rewritten each slot

A group’s grant exists only during its slot. The PC cannot connect back to students.

Group account: grp-<code> on its PC · example policy in Appendix C

HOW IT WORKS · THE SCHEDULE

The routine stays in **the Google Sheet the lab already uses**



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	PC	Mon day 06:00–17:55	Mon night 18:00–05:55	Tue day 06:00–17:55	Tue night 18:00–05:55	Wed day 06:00–17:55	Wed night 18:00–05:55	Thu day 06:00–17:55	Thu night 18:00–05:55	Fri day 06:00–17:55	Fri night 18:00–05:55	Sat day 06:00–17:55	Sat night 18:00–05:55	Sun day 06:00–17:55	Sun night 18:00–05:55
2	pc-41	MA-AI		TEST	TEST	TEST	TEST	TEST			TEST		TEST		
3	pc-43									MA-AI	TEST				
4	pc25	MA-AI													
5															
6															
7															
8															
9															

The lab's real Sheet, Routine tab · other tabs hold groups, PCs, terms and admins; Problems is written by the system

FOR STUDENTS AND ADMINS

A student needs **one sign-in and one command**

1 · GROUP

Enroll online

One member sends title, supervisor and one to five members. BRACU accounts only.

2 · ADMIN

PC and slots

An admin assigns a PC and two weekly slots in one step; they go into the Sheet.

3 · ONCE

Sign in

```
tailscale login
```

with our server’s address, or Log in in the app.

4 · IN THE SLOT

Connect

```
ssh grp-a@lab-pc-01
```

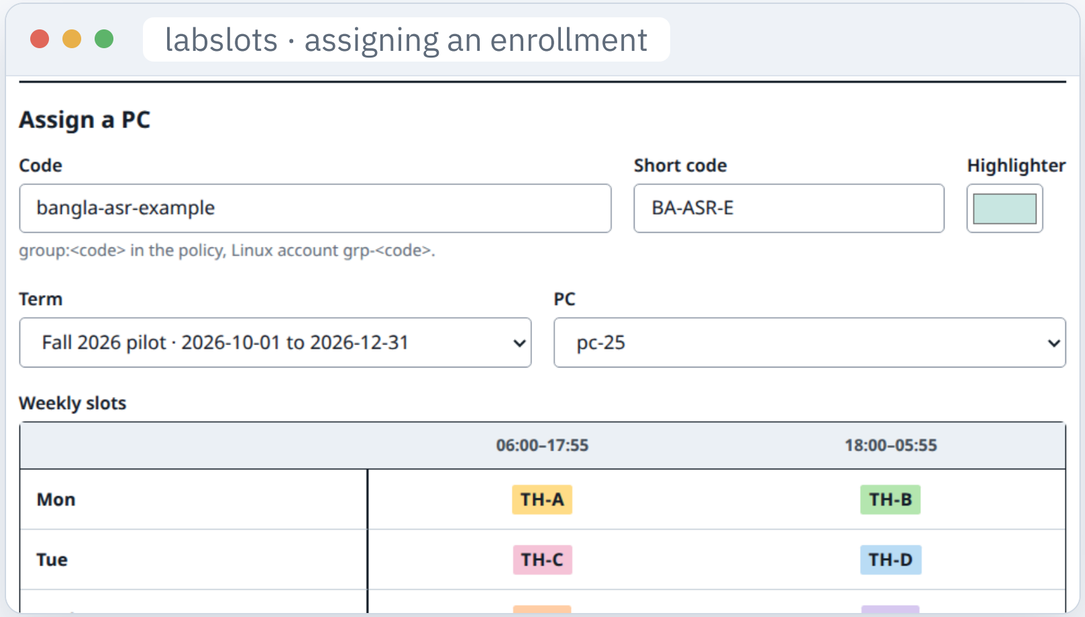
Outside the slot, the PC cannot be reached.

“Signing in proves who you are; the schedule decides what you can reach.”

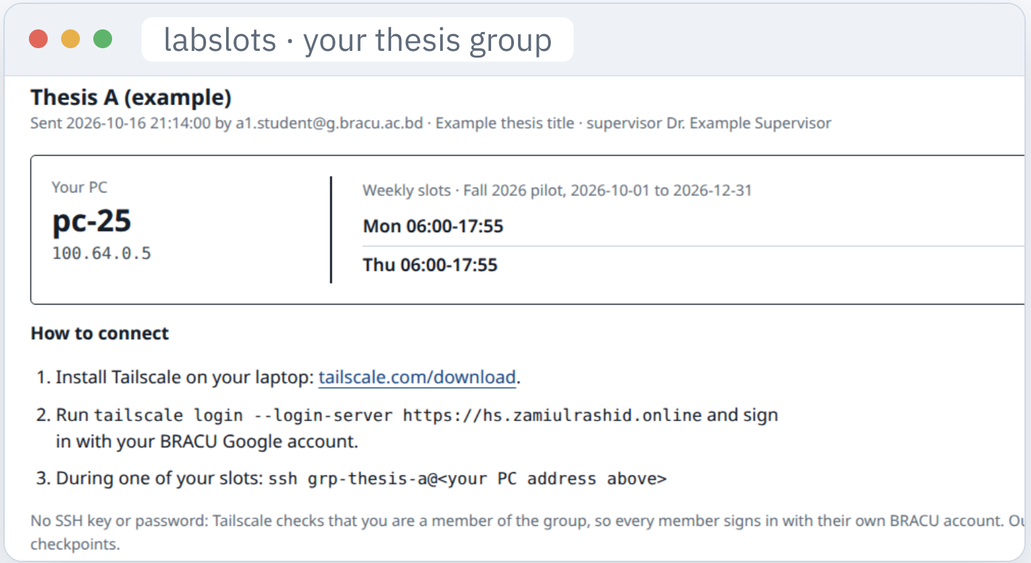
The admin desk shows the PCs in use, the routine, the enrollment queue, a policy preview and a log of every change.

HOW IT WORKS · ENROLLMENT

Enrollment replaces the Google Form, start to finish



Admin: review the application, then assign a PC and slots in one step

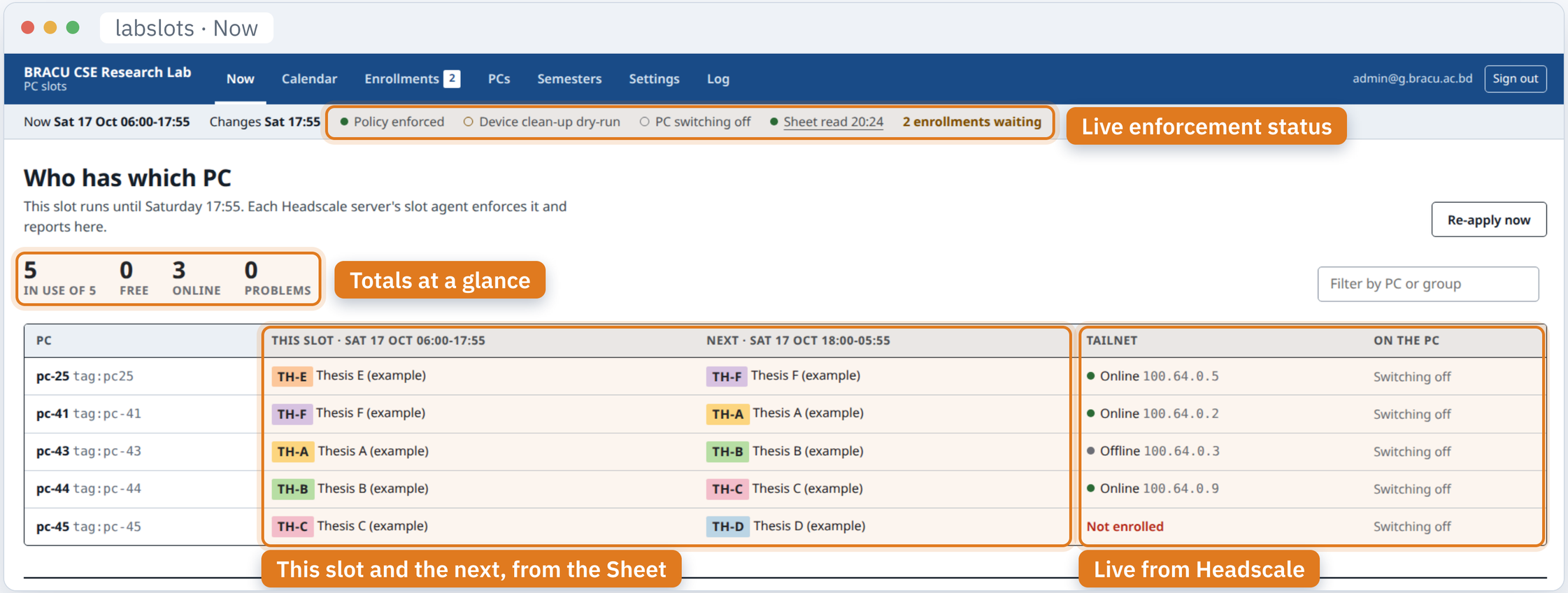


Student: the group's PC, its weekly slots and how to connect

BRACU Google accounts only; personal accounts are refused.
Every member is emailed the decision.

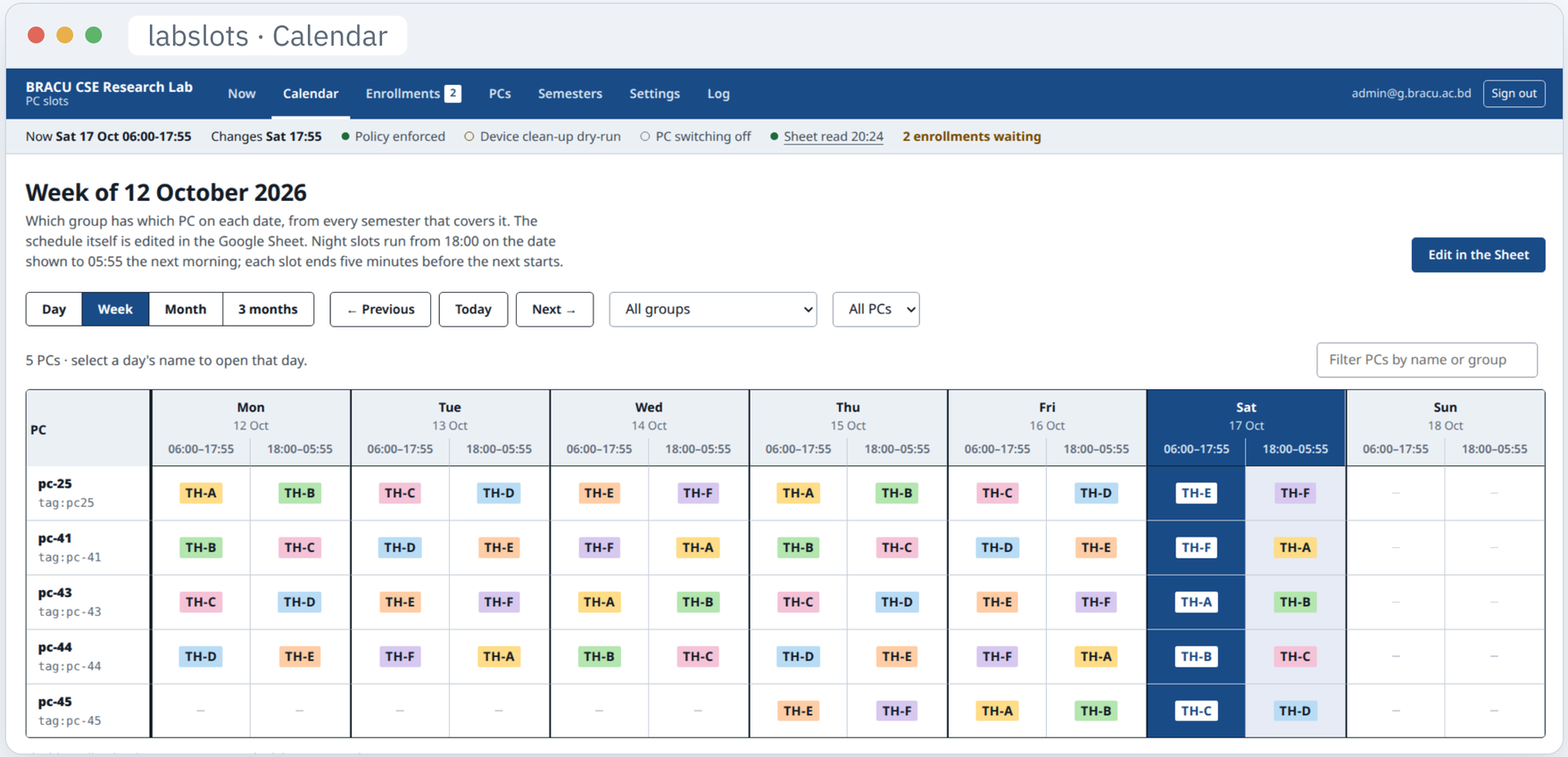
labslots website with demo data (groups marked "example")

The admin desk shows who has which PC, live



labslots website with demo data (groups marked “example”)

The calendar shows **every PC's week**, straight from the Sheet



PC	Mon 12 Oct		Tue 13 Oct		Wed 14 Oct		Thu 15 Oct		Fri 16 Oct		Sat 17 Oct		Sun 18 Oct	
	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55	06:00–17:55	18:00–05:55
pc-25 tag:pc25	TH-A	TH-B	TH-C	TH-D	TH-E	TH-F	TH-A	TH-B	TH-C	TH-D	TH-E	TH-F	–	–
pc-41 tag:pc-41	TH-B	TH-C	TH-D	TH-E	TH-F	TH-A	TH-B	TH-C	TH-D	TH-E	TH-F	TH-A	–	–
pc-43 tag:pc-43	TH-C	TH-D	TH-E	TH-F	TH-A	TH-B	TH-C	TH-D	TH-E	TH-F	TH-A	TH-B	–	–
pc-44 tag:pc-44	TH-D	TH-E	TH-F	TH-A	TH-B	TH-C	TH-D	TH-E	TH-F	TH-A	TH-B	TH-C	–	–
pc-45 tag:pc-45	–	–	–	–	–	–	TH-E	TH-F	TH-A	TH-B	TH-C	TH-D	–	–

Demo data · the dark column is the slot running now; codes and colours come from the Sheet's Groups tab

WHERE EVERYTHING RUNS

Headscale runs on the host itself; labslots runs in Docker beside it

Headscale host · Ubuntu Server 24.04 · in the lab

Caddy · TLS and the only public entry, TCP 443 · STUN, UDP 3478, goes straight to Headscale

ON THE HOST · SYSTEMD

Headscale 0.29.3

Official .deb, version pinned, upgraded by hand. SQLite and its own relay. Not in a container.

IN DOCKER

labslots website
sign-in, enrollment, admin

Slot agent
enforces the schedule

Scheduler node
agent's way onto the tailnet

Nightly backup
all three databases

Headplane
Headscale admin view

labslots ships as versioned releases

Second disk: nightly copies of the databases, keys and configuration, kept 14 days · 650 VA UPS

The machine

8-core mini PC or small desktop

16 GB RAM · 512 GB NVMe

1 GbE wired

8–15 W idle, about 11 kWh a month

The stress test used 4 cores; 8 leave room for a mass sign-in and for growth.

PART 2

What we tested

A home-lab rehearsal of the whole pilot, then a stress test at many times its size

EVIDENCE · HOME-LAB REHEARSAL, 13 OCTOBER

The rehearsal **met every target** except the share of direct connections

Measure	Target	Result	Verdict
Enrolling a lab PC	≤ 10 min	**11 s** (package installed)	**Met**
Revoking access	≤ 1 min	**3–5 s**	**Met**
Restore on a new server	≤ 2 h	**63 s**, all devices back	**Met**
Access-policy tests	all pass	**13 of 13**	**Met**
Off campus, only TCP 443 open	works	**works**, relayed, 27–37 ms	**Met**
30-minute server outage	sessions survive	direct: no loss · relayed: dropped	**Direct met**
Direct, not relayed, connections	≥ 80 %	**44–58 %**	**Below**

Below target still works: relayed connections are slower, not blocked. Giving each lab PC its own UDP port is the fix we will test on campus.

Simulated campus, firewall and off-campus clients, removed afterwards · records/ in the repository

EVIDENCE · STRESS TEST, 22 OCTOBER

On 4 cores with 871 devices, **slot changes landed within 20 s and nothing leaked**

871
devices online at peak, none removed

13–20 s
from a slot change to new rules and accounts on all 70 PCs

0 / 2,956
real logins that got in where they should not have

2 %
of one core used by Headscale between slot changes

70 simulated lab PCs, 140 groups and 700 students on three machines. Every PC ran the real setup script; every device was a real Tailscale client, checked against the schedule every 20–30 s, including logins into other groups’ accounts.

The only peak: 800 devices signing in within 3 minutes, like the first day of term, used 3.9 cores, then settled.

Run 2 of 2; full results in labslots/stress/results

PART 3

What we need

Five small things from IT, how each failure is kept safe, and the next steps

WHAT WE NEED FROM THE UNIVERSITY

We ask IT for **five small things**; the team provides everything else

#	Item	Exactly what	What it is for
1	**Public IP address**	one static IPv4, forwarded to the lab host	students reach the server
2	**Domain name**	one record under bracu.ac.bd, e.g. hs.cse.bracu.ac.bd	server and website, with a certificate
3	**Firewall ports**	inbound TCP 443 and UDP 3478; TCP 80 if allowed	control, relay and STUN
4	**Lab email**	e.g. cse-labslots@bracu.ac.bd, allowed to send	enrollment results to groups
5	**Google sign-in**	our sign-in client allowed for BRACU accounts, if restricted	students and admins sign in

Nothing else is exposed. The lab PCs, port 22 and the services behind Caddy cannot be reached from the internet.

The team provides the machine, software, certificates, setting up the PCs and running the system.

The domain name is IT’s choice; hs.cse.bracu.ac.bd is only an example

RISKS

When something fails, the system **closes access rather than opening it**

If this happens	Then
The policy cannot be built or loaded	admins-only policy; the error shows on the admin desk
The website or the Sheet is down	the agent keeps enforcing its last good copy
A mistaken edit in the Sheet	the cell keeps its last good value and is listed as a problem
Headscale restarts	direct sessions survive; upgrades are done between slots
Few direct connections	the relay still works; a port per PC improves it
A student shares an account	it still reaches only that group’s PC, in its slot
The server fails	UPS and nightly backups; a restore took 63 s

Full risk register in the written proposal

NEXT STEPS

Once IT grants network access, **the pilot can start on 15 lab PCs**

1

Network access

IT provides the IP address, domain name, ports and lab email.

[date]

2

Server on campus

We install the host in the lab and move our running setup onto it.

[date]

3

15 PCs and groups

Each PC is prepared with one script; the term’s groups enroll.

[date]

4

One term, measured

We run it, measure against the targets and report to the department.

[date]

AFTER THE PILOT Remote desktop over the tailnet (RustDesk or xrdp) under the same policy, replacing AnyDesk. Designed, not built yet.

CONCLUSIONS

Booked slots can be enforced, and the system to do it is ready

1. **Only the group in its slot reaches its PC**, as its own account, with BRACU sign-in and no passwords.

2. **It is built and tested:** 871 devices on 4 cores, changes within 20 s, no leaks in 2,956 checks.

3. **The pilot needs little from IT:** one IP address, one domain name, two ports and a lab email account.

Zamiul Rashid · Akid Anis · Tanzim Ahmed

Supervisor Md. Golam Rabiul Alam · Co-supervisor Mollah Md. Saif · [contact email]

`github.com/Zamiul-rashid/Headscale`

APPENDIX

References

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APPENDIX A · DEVICES

A 15-PC pilot means **91 to 916 devices**, well inside what we tested

Situation (15 lab PCs and 1 scheduler included)	1 device per student	2 devices per student
One slot, all 15 PCs in use, 5 students per group	**91**	**166**
Devices kept all week (6 groups per PC, 90 groups)	**466**	**916**

A device is not a user: a student with a laptop and a phone is two devices. Devices stay registered; access follows the policy, not the device list.

A day after a student leaves every group, their devices are removed. An optional switch also removes extra devices; it starts as a dry run.

Stress test held 871 devices on 4 cores (slide 18)

APPENDIX B · REMOTE DESKTOP

AnyDesk goes around the policy, so **remote desktop moves onto the tailnet**

Today: AnyDesk

Connects through AnyDesk’s own servers, not our tailnet, so our policy cannot control it. Its free licence does not cover lab use.

Planned: over the tailnet

Unattended AnyDesk turned off on enrolled PCs. RustDesk in direct mode, or xrdp, on a port added to the group’s slot.

Status: designed, not built or tested yet. The same policy will then control remote desktop exactly as it controls SSH.

APPENDIX C · THE POLICY

The policy is **rewritten at every slot change**; this is Monday 06:00

```
"groups": { "group:thesis-a": ["s2101@g.bracu.ac.bd", "s2102@g.bracu.ac.bd", …] },
"grants": [
  { "src": ["group:admins"],   "dst": ["tag:pc-01", …], "ip": ["*"] },
  { "src": ["tag:scheduler"],  "dst": ["tag:pc-01", …], "ip": ["tcp:22"] },
  // this slot, Mon 06:00–17:55: pc-01 shared with Group A
  { "src": ["group:thesis-a"], "dst": ["tag:pc-01"],    "ip": ["tcp:22"] } ],
"ssh": [
  { "action": "accept", "src": ["group:thesis-a"], "dst": ["tag:pc-01"],    "users": ["grp-thesis-a"] },
  { "action": "accept", "src": ["tag:scheduler"],  "dst": ["tag:pc-01", …], "users": ["labsched"] } ]
```

The orange lines are the only part that changes between slots. If the policy cannot be built, the agent loads the admins-only version instead.

Headscale policy format (grants and ssh rules)